THE ONTOLOGY IMPERATIVE

Trustworthy Agentic AI for Boards and Executive Search

Chief Data Officer

Governance Mandate, Reference Job Description

Version 1.1 · Modes 1 and 2 · 13 June 2026

Role summary

The **Chief Data Officer (CDO)** is the executive accountable for the structural governance of agentic AI in the enterprise. The role exists because AI agents act under authority the organisation grants and reason over data the organisation holds; the structural authority to govern what these agents are permitted to know, do, and decide cannot be distributed across IT, business units, and vendor platforms without creating an accountability gap that no other executive can close.

This reference job description covers the CDO role at organisational maturity Modes 1 (**Governed Infrastructure**) and 2 (**Delegated Governance**), the two maturity stages where the CDO holds governance accountability without inheriting operational accountability for the processes that agents serve. The Mode 3 variant (**Operational Accountability** for a demanned division) is a categorically different role and is described in a separate artefact.

Reporting and scope

Reports to	Chief Executive Officer. The standards-setting authority cuts across Data Owners, the CIO, the COO, and Process Owners; only the CEO has authority over all of those functions simultaneously. There is no CAIO.
Direct reports	Heads of the seven capability groups (see the organisation under the CDO, page 2).
Cross-functional partners	Data Owners (across Data Domains); Process Owners (across business processes); CIO; COO; General Counsel; Chief Risk Officer.
Board interface	Member of, or standing presenter to, the board audit and risk committee. Standard-of-care attestations on agent governance feed board oversight.
Mandate horizon	Indefinite. The CDO is a permanent enterprise function, not a transformation role.

Structural authority

The CDO mandate carries four structural authorities that cannot be partially delegated. Together they constitute the fourth line of defence: deterministic enforcement that fires before agent action, operating at machine tempo alongside the three traditional lines (operational management, risk and compliance, internal audit), which operate at human tempo. Independent assurers, for example an accredited certification body, attest the fourth line; they are not the fourth line.

Meaning and connectivity	Owns the enterprise ontology on open standards (W3C OWL), the guardrails (SHACL), provenance (PROV-O), agent contracts (ODRL), and the connectivity protocols (MCP, A2A). Sets the meaning capability against which all agents reason and the protocols by which they connect. Capabilities L3 to L8 of the Agentic AI Capability Stack™.
Two structural gates	Operates the **Ingestion Gate** (L2/L3): Data Owner attests data meets agent-readiness; CDO approves admission to the meaning capability. Operates the **Deployment Gate** (L8/L9): CDO attests governance content is complete and scoped; Process Owner approves operational acceptance.
Halt authority	Retains unilateral authority to halt any agent for governance breach, parallel to the Process Owner's authority to halt for operational failure. Reinstatement requires both signatures.
Audit chain	Owns the provenance capability (PROV-O at L7) that produces machine-readable, reconstructable audit evidence for every agent decision. The structural answer to the standard-of-care question after any agent-driven loss.

The organisation under the CDO

The CDO office is a governance function, not a data team with a new title. Seven capability groups report to the CDO. Four build and operate the meaning capabilities (L3 to L9). One runs assurance across the audit chain, structurally independent from Governance Operations so the gate operators are not auditing themselves. Two work the data foundation and the Ingestion Gate (L1 to L3): one governs what data means and stewards it; the other assures it is fit for agents to reason over.

Capability group	What it does	Capability
Ontology Engineering	Formal ontologies (OWL), controlled vocabularies (SKOS). Aligned to BFO, POSC Caesar, CFIHOS, FIBO.	**L3**
Knowledge Graph Architecture	Organisational knowledge graph (RDF, SPARQL). Grounded in the L3 ontology.	**L4**
Governance Operations	Agent contracts (ODRL), guardrails (SHACL), provenance (PROV-O). Operates both gates. Continuous L9 runtime monitoring supporting halt authority.	**L5–L9**
Agent Integration	Agent connectivity protocols (MCP, A2A). Deployment governance. Uniform protocol access.	**L8–L9**
Governance Assurance	Audit-chain analysis across agents and over time. Drift detection, asymmetry surfacing, near-miss pattern reporting. Independent from Governance Operations; feeds board oversight.	**L5–L11**
Data Governance & Stewardship	Data policy, ownership, semantic compatibility, and provenance standards at source. Dotted-line to Data Stewards.	**L1–L2**
Data Quality Management	Sets agent-readiness quality standards (completeness, conformance, freshness) and runs the fitness checks the Ingestion Gate attestation rests on. Guards meaning against agent write-back degradation across the chain. Works with Data Governance towards Data Owners, Stewards, and custodians.	**L1–L3**

Required profile

Senior leadership trajectory totalling 15+ years across data, governance, and integrated operations, including executive-level accountability (Head of, VP, or C-1) on enterprise-scale data or governance programmes. Track record of operating board-facing governance functions in regulated, high-liability, or safety-critical environments.

Architectural fluency in the W3C semantic stack (OWL, SHACL, PROV-O, RDF, SPARQL). This is not a delegable competency. The CDO must be able to engage at architectural depth with ontology engineers, vendor architects, and external auditors; reading semantic specifications and reasoning about constraints is part of the day-to-day role.

Cross-functional executive presence. Ability to engage with Process Owners, Data Owners, CIO, COO, General Counsel, and Chief Risk Officer as peer, not as adviser. The mandate carries veto authority at the two structural gates; the role-holder must be able to exercise that authority against operational pressure when required.

Vendor governance experience. Demonstrated ability to negotiate semantic, enforcement, execution, and jurisdictional sovereignty into platform contracts as deal-stoppers. Familiarity with the contractual criteria that prevent sovereignty collapse to vendor platforms.

Equivalent senior backgrounds

The first generation of agentic-AI-capable CDOs will come from a range of adjacent senior leadership backgrounds, not from prior tenure in a role that the architecture has only recently made coherent. Recognised entry paths include Heads of Data Office, VPs of Data and Analytics with cross-functional accountability, Directors of Integrated Operations with data architecture remits, programme leaders on enterprise data or governance transformation, senior data architects with explicit governance accountability, and CDOs from adjacent sectors making lateral transitions. What unifies them is direct accountability for enterprise data architecture at scale combined with architectural depth in semantic technologies. **Prior CDO tenure is one path among these, not a prerequisite.**

Strongly preferred

Architectural depth on at least one production knowledge graph or ontology deployment at enterprise scale. Experience with insurance and audit frameworks for AI liability (D&O and professional indemnity calibrations). Industrial domain depth in the hiring organisation's sector. Familiarity with offshore safety, financial controls, or other dual-key operational governance conventions (parallel to the joint-custody pattern between CDO and Process Owner at L9).

Scaling from Mode 1 to Mode 2

The CDO role scales materially in scope, team size, and KPI orientation between Modes 1 and 2, though the structural authorities and reporting line do not change. The two structural gates apply at every mode; only the delegation scales. A CDO who builds successful Mode 1 infrastructure evolves into the Mode 2 role; the role does not change identity.

Dimension	Mode 1: Governed Infrastructure	Mode 2: Delegated Governance
Scope	A few agents inside a single business unit. Foundational ontology under construction. The CDO defines standard governance contracts and holds both gates; the Process Owner selects from pre-approved templates and retains operational accountability for outcomes.	One fully agentic workflow, end to end, the pattern extending to further workflows as they mature. The Process Owner formally delegates technical governance of the agentic portion to the CDO by charter or board resolution, and retains operational liability for outcomes.
Team size	10 to 15. Ontology Engineering and Governance Operations primary; other capability groups in formation.	25 to 40. All seven capability groups operational. Governance Assurance structurally independent.
Primary KPIs	Number of vetted agent templates; Ingestion Gate throughput; Deployment Gate approval rate; zero unauthorised deployments.	Workflow governance maturity; charter completeness; halt-and-restart cycles; provenance audit coverage; insurance posture.
Time allocation	~60% infrastructure (ontology, knowledge graph, gates); ~30% governance operations; ~10% board and C-suite engagement.	~40% governance operations; ~30% Process Owner strategic engagement; ~20% board and C-suite; ~10% infrastructure evolution.
Transition	18 to 24 months typical to mature into Mode 2, contingent on agent footprint growth.	Indefinite for many processes. Mode 3 emerges only where divisions become demanned.

Risk and liability profile

The CDO Governance Mandate carries explicit **governance accountability**, not operational accountability. The CDO answers for whether structural enforcement was complete, scoped, and operating at the time of any agent-driven loss event; the CDO does not answer for the operational outcomes that follow a gate fire. Where Article 26 of the EU AI Act, the 2024 EU Product Liability Directive revision, US *Caremark* precedents, or similar frameworks elsewhere create operator-level liability for AI-driven outcomes, that liability flows to the Process Owner under post-gate joint custody, not to the CDO.

The CDO is accountable for **evidencing that the structural controls were in place** at the time of the loss event: the governance-content attestation at deployment, the provenance audit chain, and the operation of the halt authority. This evidence must be reconstructable from the audit chain (PROV-O) without reliance on the vendor. D&O coverage and professional indemnity calibrations for the role should reflect this scope. The role is structurally insurable; an uninsurable CDO mandate is a sign the mandate is not yet structurally enforceable.

About this artefact

This reference job description describes the Chief Data Officer role at organisational Modes 1 (Governed Infrastructure) and 2 (Delegated Governance). The Mode 3 variant (Operational Accountability for a demanned division) is a categorically different role and is described in a forthcoming separate artefact.

Canonical references: the Agentic AI Capability Stack™, Version 3.1 (capability and governance framework); The CDO Office: Mandate, Organisation, and Operating Modes, Version 1.2 (full mandate); Article 3b for the framework argument and Article 3c, When the Container Fails, for the sovereignty and liability framework, both at theontologyimperative.substack.com. All artefacts available at fredericverhelst.com/TOI-library.